Harikrishnan Sivaraman
251901051

Ex. No.: 1

CAPTURE FLAGS-ENCRYPTION CRYPTO 101

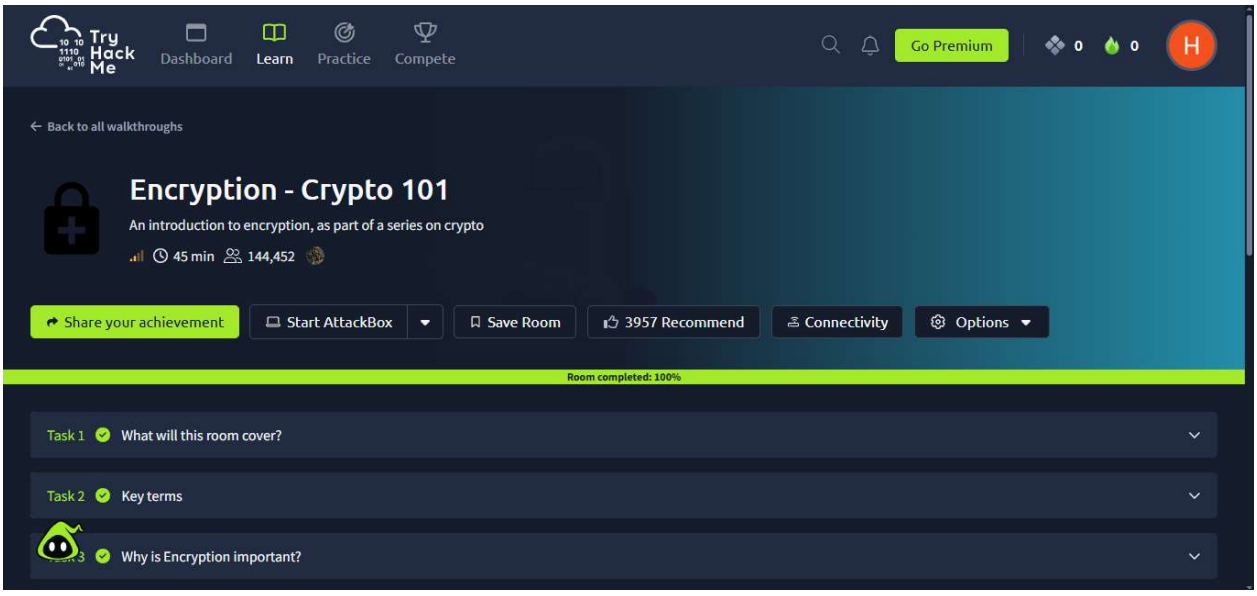
Aim:

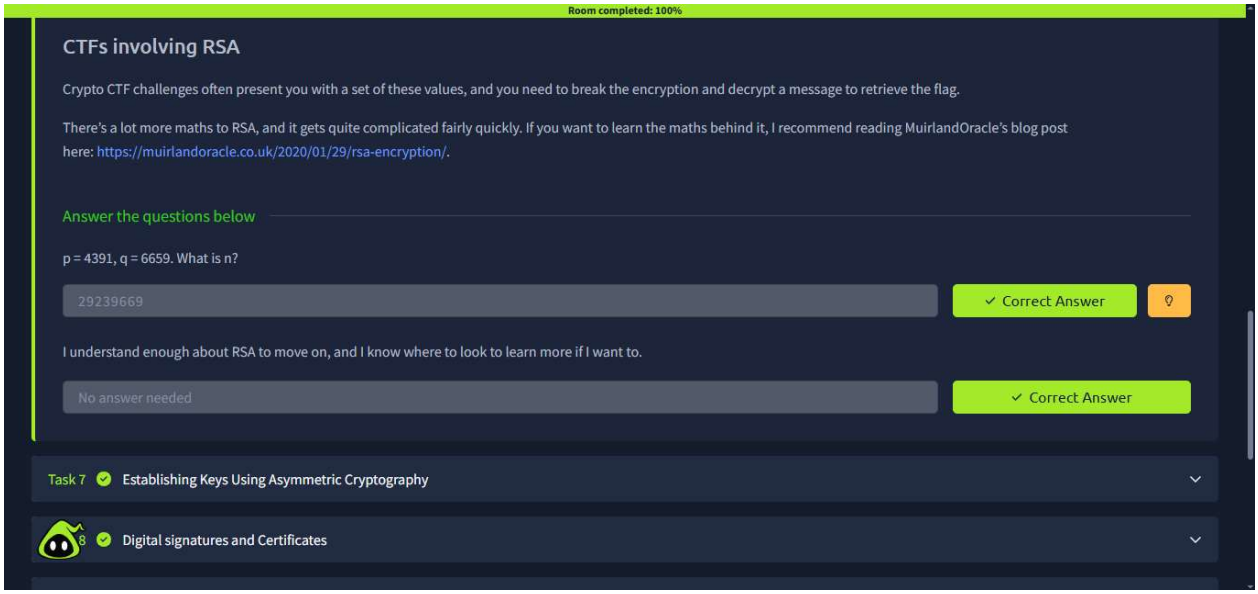
To capture the various flags in Encryption Crypto 101 in TryHackMe platform.

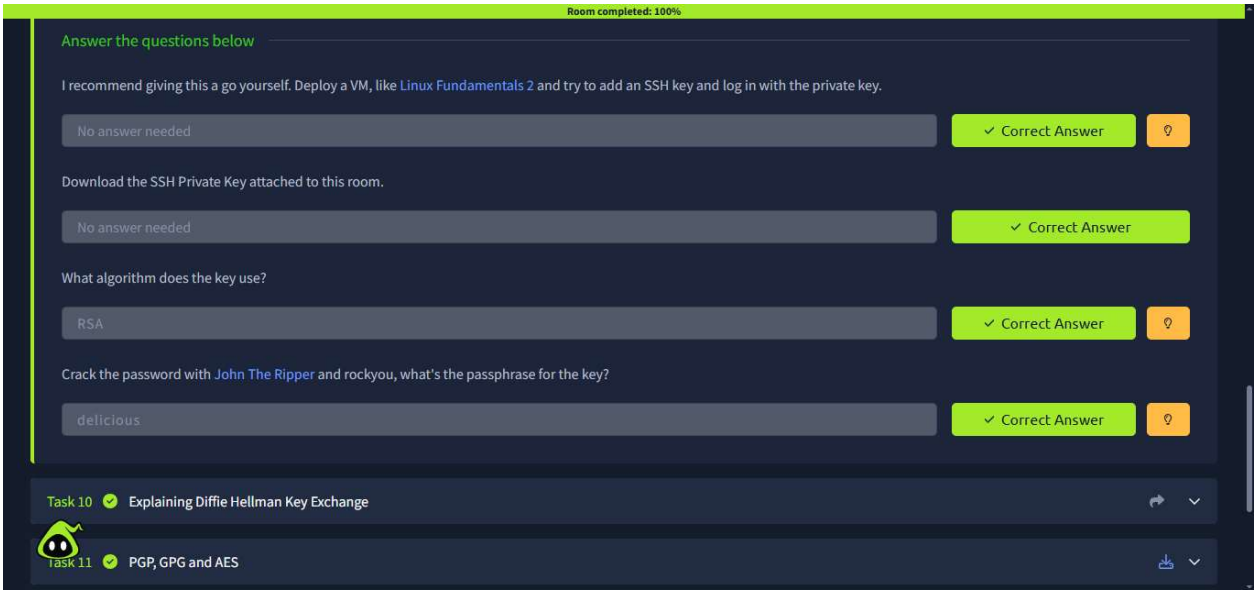
Algorithm:

1. Open the Passive Reconnaissance lab on TryHackMe using the provided link.
2. Launch the AttackBox to start the Kali Linux environment.
3. Solve the mathematical calculations involved in RSA cryptography.
4. Identify the Certificate Authority (CA) that issued the HTTPS certificate for tryhackme.com.
5. Generate an SSH public and private key pair using ssh-keygen and use it for SSH authentication.
6. Decrypt the provided GPG-encrypted file and determine the hidden secret word.

Output:




Room completed: 100%
CTFs involving RSA
Crypto CTF challenges often present you with a set of these values, and you need to break the encryption and decrypt a message to retrieve the flag.
There's a lot more maths to RSA, and it gets quite complicated fairly quickly. If you want to learn the maths behind it, I recommend reading MuirlandOracle's blog post here: https://muirlandoracle.co.uk/2020/01/29/rsa-encryption/.
Answer the questions below
p = 4391, q = 6659. What is n?
29239669
✓ Correct Answer
I understand enough about RSA to move on, and I know where to look to learn more if I want to.
No answer needed
✓ Correct Answer
Task 7 Establishing Keys Using Asymmetric Cryptography
8 Digital signatures and Certificates


Room completed: 100%
Answer the questions below
I recommend giving this a go yourself. Deploy a VM, like Linux Fundamentals 2 and try to add an SSH key and log in with the private key.
No answer needed
✓ Correct Answer
Download the SSH Private Key attached to this room.
No answer needed
✓ Correct Answer
What algorithm does the key use?
RSA
✓ Correct Answer
Crack the password with John The Ripper and rockyou, what's the passphrase for the key?
delicious
✓ Correct Answer
Task 10 Explaining Diffie Hellman Key Exchange
Task 11 PGP, GPG and AES

```
root@ip-10-10-18-189:~# gpg message.gpg
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: encrypted with 1024-bit RSA key, ID 2A0A5FDC5081B1C5, created 2020-06-30
"TryHackMe (Example Key)"
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: encrypted with 1024-bit RSA key, ID 2A0A5FDC5081B1C5, created 2020-06-30
"TryHackMe (Example Key)"
```

Result:

Thus, all the required flags were successfully obtained while completing the Encryption Crypto 101 room on the TryHackMe platform.